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ANNEX 2

ECOLOGICAL AND SOCIO-ECONOMIC CRITERIA OF THE NORTH-WESTERN MEDITERRANEAN SEA PARTICULARLY SENSITIVE SEA AREA (NW MED PSSA)*

1 Introduction

1.1 The NW Med PSSA covers a perimeter corresponding to the eastern boundary of the Pelagos Sanctuary and to the west to the Spanish cetacean migration corridor. These are two specially protected areas of Mediterranean importance (SPAMI) established under the Barcelona Convention and dedicated to cetaceans including over 230 EU Natura 200 0 sites. The area fully or partly overlaps, under the Convention on Biological Diversity (CBD) framework, the two ecologically or biologically significant marine areas (EBSA) and three important marine mammal areas (IMMA) identified by the International Union for Conservation of Nature (IUCN) Marine Mammal Protected area task force. This perimeter also includes most of the Strait of Bonifacio PSSA.

Physical features

1.2 The North-Western Mediterranean portion of the basin is characterized by the rapid

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of Nature (IUCN) Marine Mammal Protected area task force. This perimeter also includes most of the Strait of Bonifacio PSSA.

Physical features

1.2 The North-Western Mediterranean portion of the basin is characterized by the rapid plunge of its coasts towards the deep -sea (up to 2,000 m in some area) in proximity of the main islands (Corsica and Sardinia) and off the Ligurian coasts and most of th e Provence-Alpes-Côte d'Azur's and Catalonia's coasts. The continental shelf is developed off Tuscan coasts (including all around the Tuscan archipelago) and Valencian coasts, with a maximum extension (about 100 km wide) within the study area in the Gulf of Lion (Occitanie).

1.3 Another notable feature of the North-West Mediterranean seabed is that it shows one of the highest densities of canyons globally, veritable submarine valleys present on the oceanic slope, generally between 300 and 600 m deep. Canyons are usually define d from the border of the continental shelf, having their "head" beginning at -200 m deep, and finishing at the bottom of the oceanic bed at -2,000 m deep.

1.4 The Mediterranean is an evaporation basin: precipitation and river inputs do not

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of the continental shelf, having their "head" beginning at -200 m deep, and finishing at the bottom of the oceanic bed at -2,000 m deep.

1.4 The Mediterranean is an evaporation basin: precipitation and river inputs do not compensate for evaporation. This water deficit is made up by Atlantic water entering the surface through the Strait of Gibraltar. Less salty and therefore less dense than the Mediterranean water, this water will remain on the surface and determine the surface circulation.

1.5 The surface currents have a complex organization, particularly around Corsica.
The main horizontal marine currents have a so -called cyclonic direction (counterclockwise).
The areas where they reach a higher intensity in our study area, i.e. an average annual speed of more than 0.25 m per second, are the Ligurian Sea and the Tyrrhenian Sea, east of Bonifacio (Corsica). Seasonal trends show an increase in speed during the summer and autumn.

1.6 Upwelling phenomena, vertical currents that allow deep water to rise to the surface, are due to a combination of horizontal currents with the wind and can be influenced by the

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1.6 Upwelling phenomena, vertical currents that allow deep water to rise to the surface, are due to a combination of horizontal currents with the wind and can be influenced by the presence of submarine canyons. The Ligurian Sea and the northern Tyrrhenian Sea are the most exposed to this phenomenon. In spring, the increase in the temperature of the marine waters leads to a vertical stabilization of the water masses. Thus, marine currents play a very important role in the functioning of ecosystems: through their associated horizontal and vertical movements, they accompany the export of organic matter from the coast to the open sea.

* The text in this annex is drawn from information provided by France, Italy, Monaco and Spain in document MEPC 79/10. All references used in this resolution are set out in the annex to document MEPC 79/10.

1.7 The North-Western Mediterranean Sea is one of the world's 10 biodiversity hot spots,

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1.7 The North-Western Mediterranean Sea is one of the world's 10 biodiversity hot spots, and although it represents only 1% of the total surface area of the oceans, it is home to around 10% of the world's recorded species. The area therefore simultaneously evidences multiple criteria for PSSA designation: critical habitat, dependency, diversity, productivity,

spawning or breeding grounds, fragility, biogeographic importance, social or economic dependency, and research and education. These crit eria, which are described below, demonstrate the global importance of the area, they have also been considered thoroughly under a number of policy frameworks on natural heritage and socio-economic elements of the marine environment, including under the CBD and Barcelona conventions, and European Union policies (e.g. Marine Spatial Planning, Marine Strategy Framework Directive, Common Fishery Policy, Habitats Directive and the General Fisheries Commission for the Mediterranean (GFCM)).

2 Ecological criteria

Uniqueness or rarity

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2 Ecological criteria

Uniqueness or rarity

2.1 The North-Western Mediterranean is part of a semi-enclosed sea with a high rate of endemism. The vast majority of its biological populations are composed of Mediterranean subpopulations, genetically isolated from the Atlantic populations and the others.

Critical habitat

2.2 The ecological and biological significance of the PSSA is supported by the existence of two areas listed under the CBD framework of the EBSAs, which are overlapping it:

.1 the North-Western Mediterranean Benthic Ecosystems; and

.2 the North-Western Mediterranean Pelagic Ecosystems.

In addition, over two thirds of the PSSA are covered by the "North -Western Mediterranean Sea, Slope and Canyon System", the "The Shelf of the Gulf of Lion" and the "Western Ligurian Sea and Genoa Canyon" IMMAs, identified by the IUCN Marine Mammal Protecti on Working Group. Moreover, the PSSA includes a candidate IMMA (the "Central Tyrrhenian Sea IMMA")

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Sea and Genoa Canyon" IMMAs, identified by the IUCN Marine Mammal Protecti on Working Group. Moreover, the PSSA includes a candidate IMMA (the "Central Tyrrhenian Sea IMMA") and an Area of Interest (the Tuscan Archipelago), which could soon become IMMAs. It is also adjacent to the "Balearic Islands Shelf and Slope IMMA", off the southern coasts of Balearic Islands, a critical habitat for the Mediterranean sperm whale. It also includes the Pelagos Sanctuary for marine mammals.

2.3 These areas have a set of geomorphological and oceanographic features that favour productivity levels of extraordinary biological and ecological importance for the region.
In particular, the PSSA area overlaps important habitats for the endangered Mediterranean fin whales (Balaenoptera physalus), the endangered sperm whales (Physeter macrocephalus), the

vulnerable Cuvier's beaked whales (Ziphius cavirostris), the Habitats European Directive Annex II bottlenose dolphins (Tursiops truncatus) and the endangered Risso's dolphins (Grampus griseus) (ACCOBAMS 2022). All cetacean species are also listed in Annex IV of the Habitats, Fauna and

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bottlenose dolphins (Tursiops truncatus) and the endangered Risso's dolphins (Grampus griseus) (ACCOBAMS 2022). All cetacean species are also listed in Annex IV of the Habitats, Fauna and Flora European Directive 92/43/EEC (animal and plant species of Community interest that require strict protection). These species are included in the IUCN red list.

2.4 The preservation of cetaceans is a necessity in terms of maintaining the ecological balance in the Mediterranean Sea and contributes to the mitigation of climate change (Roman et al. 2014) their economic value must also be considered, as cetaceans play a major role in the development of tourism in the area. Finally, from the point of view of biodiversity, some of the Mediterranean cetacean subpopulations are genetically isolated from Atlantic populations and the others (e.g. fin and sperm whales), which gives them a unique value.

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2.5 Numerous studies have attempted to define the habitat of cetaceans and distinguish the presence of different species by physical and hydrological factors such as surface water temperature and the different water masses present, topographical features, and currents. The presence of cetaceans is often dependent on the distribution of the prey they feed on. The continental slope is the preferred habitat of species with a specialized diet composed mainly of cephalopods: the sperm whale, the Cuvier 's beaked whale, the long -finned pilot whale and the Risso's dolphin; the great abyssal plain is the preferred habitat of the fin whale. The bottlenose dolphin prefers waters to the continental shelf, usually within the 100 m isobath.

2.6 The PSSA is frequented by several species of cetaceans, eight of which (fin, sperm, Cuvier's beaked and long -finned pilot whales, Risso 's, bottlenose, striped and common dolphins) are regularly present all year round.

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dolphins) are regularly present all year round.

2.7 The importance of this area for fin whales is clear: the estimated abundance of this species within the PSSA represents about 67% of the whole Mediterranean population (ACCOBAMS 2021). Concerning the sperm whale, compared to the total Mediterranean estimate of about 1,400 individuals (ACCOBAMS 2021), the estimate in half of the PSSA (the whole Pelagos Sanctuary and French waters; Laran et al., 2017) was between 300 and 600 individuals, with higher numbers in winter. The predicted distribution of these t wo species is shown in figures 2 and 3.

Figure 2: Above: Fin whale predicted densities (summer data: 1999-2016) (Mannocci et al., 2018);
Below: Fin whale predicted densities (summer 2018) (ACCOBAMS, 2021).

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Figure 3: Sperm whale sightings and acoustic detections (ASI 2018, white squares and red/orange circles), overlaid on a predictive density map from Mannocci et al., 2018 (yellow = highest probability, blue = lowest probability) (ACCOBAMS, 2021).

2.8 To date, there is no fine-scale mapping of the preferential habitats of these cetacean species for the entire North-Western Mediterranean basin that could be used to guide a zoning approach. Thus, the identification of areas with higher risk of collision between ships and sensitive species (the fin whale and the sperm whale) within the PSSA is complex.

2.9 The latest cetacean research campaigns in the Mediterranean carried out as part of the ACCOBAMS Survey Initiative (ASI) have confirmed the knowledge on preferential presence of fin whales within the PSSA (figure 2), particularly from off the Gulf of Li on to the coastal and offshore waters off Catalonia. Concerning the offshore areas, this may be linked

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presence of fin whales within the PSSA (figure 2), particularly from off the Gulf of Li on to the

coastal and offshore waters off Catalonia. Concerning the offshore areas, this may be linked to the presence of cyclonic eddies that are the main reason for the high productivity of the area, as canyons play more a local role.

2.10 The North-Western Mediterranean has one of the highest densities of marine canyons recorded globally and regionally, which probably strongly contributes to make it highly productive (see section 2.13). Concerning the Spanish offshore and coastal areas, this has been recently confirmed as a core-feeding habitat for fin whales, especially in shallow coastal waters, through satellite tagging. Interestingly, these coastal waters coincide with areas of higher density within the North -West Mediterranean for European sardine s (Sardina pilchardus) and European anchovies (European anchovy). Their occurrence extends to the edge of the continental shelf and their distribution generally overlaps, although sardines are distributed closer to the coast and reach larger sizes (EC et al,. 2020).

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to the edge of the continental shelf and their distribution generally overlaps, although sardines are distributed closer to the coast and reach larger sizes (EC et al,. 2020).

2.11 Concerning the sperm whale, the predicted distribution by Mannocci and colleagues (2018) (figure 3) shows higher densities in the area within the PSSA between the Balearic Islands and the Spanish continental coast.

2.12 A synthesis of distribution of both species has also been carried out for the Pelagos Sanctuary and adjacent waters (Laran et al., 2012). Based on multiple data sets over 15 years, gathe